UNIT 1

CELL MEMBRANES



S. J. Singer and G.L.Nicolson

The Fluid mosaic model was first proposed by S.J Singer (American Cell Biologist) and G L. Nicolson (American Biochemist) in 1972 to describe the structure of cell membrane. They published a paper entitled "The structure of cell membrane" in cell biology journal. In this model cell membrane consists of two phospholipid layers. Each phospholipid macromolecule is itself composed of a hydrophilic head and hydrophobic tail. The hydrophilic heads are attracted to water whereas the hydrophobic tails are repelled by water. This attraction and repulsion gives the stable structure to lipid bilayer.

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Learning Objectives

After studying this unit the students will be able to

- discuss the structure and composition of a biological cell membrane
- explain the biomemebrane models
- describe the transport of substances across the membrane
- examine the properties of membrane
- explain the various buffer systems and functions

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